

Magic Item Creation Feats — Implementation Plan

D&D 3.5e Prototype • Pre-Combat Crafting Workshop

Comprehensive design for all 8 PHB item creation feats

Target: `/home/ubuntu/dnd35prototype/Assets/Scripts/Crafting/`

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1. Overview

What Is Item Creation?

Item creation feats allow spellcasting characters to craft magic items during downtime, spending gold, XP, and time instead of finding items as loot. This is one of the most impactful subsystems in D&D 3.5e because it lets players convert surplus resources into precisely the items they need.

Why It Matters for the Prototype

Impact Area	Effect
Economy	Players spend gold + XP to create items at half market price
Power Level	Custom item selection is extremely powerful — often considered the strongest feat category
Time Gating	Crafting costs real in-game days — prevents unlimited creation between encounters
Wizard Identity	Wizards get item creation feats as bonus feats (levels 5/10/15/20)
Scroll Universality	Scribe Scroll is free for all spellcasters — every caster can make scrolls

Core Formula Summary

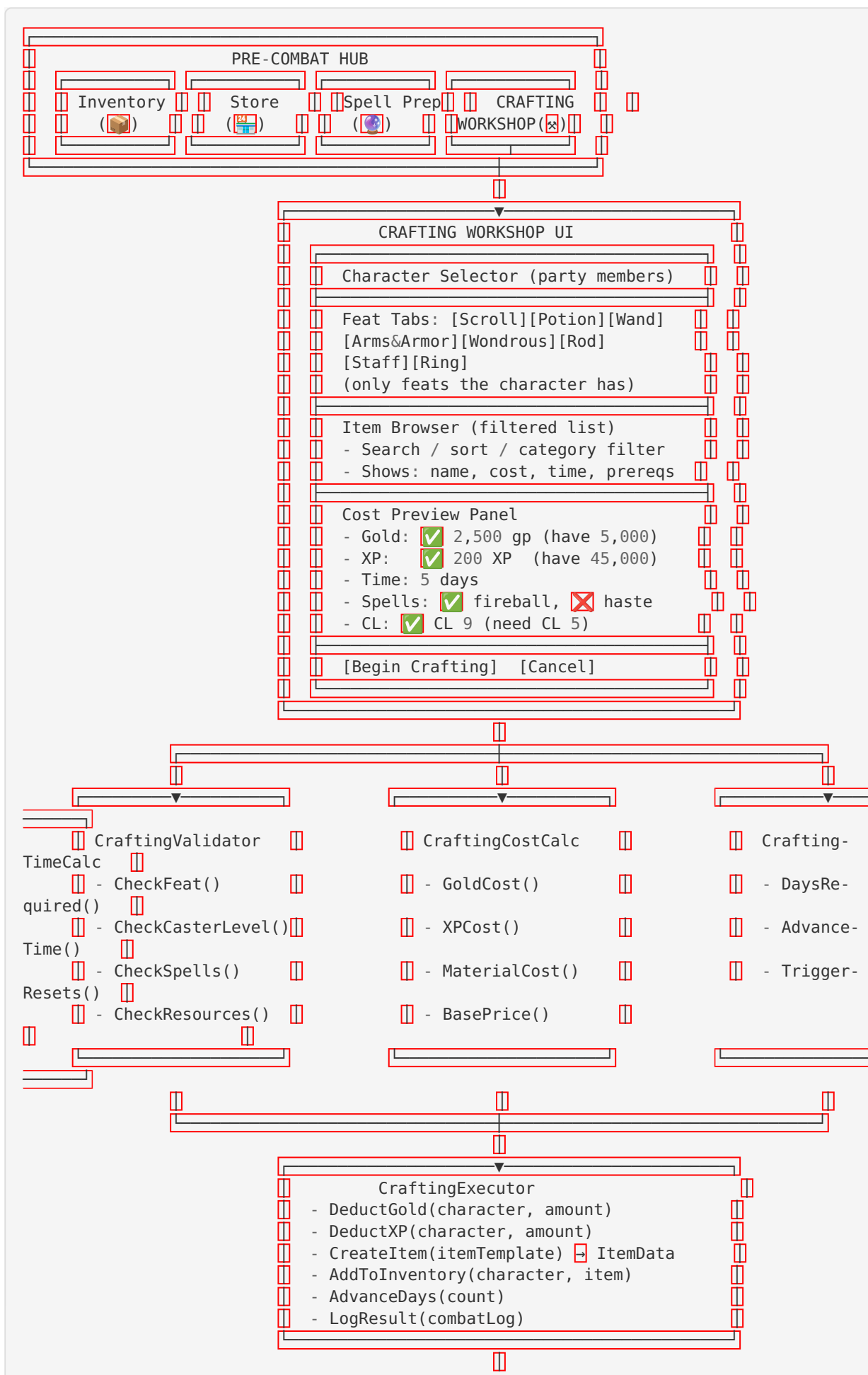
Gold Cost = Market Price / 2
 XP Cost = Market Price / 25
 Time = 1 day per 1,000 gp of base price (minimum 1 day)

Design Principles

1. **Pre-Combat Only** — Crafting happens in the Pre-Combat Hub menu, never during combat
 2. **Immediate Resolution** — Time “fast-forwards”; no real-time waiting
 3. **Full Validation** — Check feat, caster level, spells, gold, XP before allowing crafting
 4. **Leverage Existing Databases** — All craftable items already exist in our item databases
 5. **No Level Loss** — Prevent XP expenditure that would drop the character below their current level
-

2. System Architecture

Component Diagram



EXISTING SYSTEMS		
CharacterStats	XP, gold, feats, level	
SpellcastingComponent	known spells	
ItemDatabase	item templates	
RodDatabase/RingDatabase/etc.	categories	
Inventory	add items to character	
GameManager	time/rest resets	

New Files to Create

```
Assets/Scripts/Crafting/
├─ CraftingFeatType.cs           # Enum for 8 creation feats
├─ CraftableItemDefinition.cs    # Template for a craftable item
├─ CraftableItemRegistry.cs     # Registry of all craftable items per feat
├─ CraftingCostCalculator.cs     # Gold, XP, base price calculations
├─ CraftingTimeCalculator.cs     # Day calculations, time advancement
├─ CraftingValidator.cs         # Full prerequisite validation
├─ CraftingExecutor.cs          # Execute crafting (deduct costs, create item)
├─ CraftingProject.cs           # Data class for an in-progress crafting project
├─ CraftingWorkshopUI.cs        # Main crafting menu UI
├─ CraftingItemBrowserPanel.cs  # Item selection/browsing sub-panel
├─ CraftingCostPreviewPanel.cs  # Cost/prereq preview sub-panel
└─ CraftingConfirmationDialog.cs # "Are you sure?" confirmation
```

Modified Files

```
Assets/Scripts/UI/PreCombatHubUI.cs    # Add "✕ Crafting Workshop" button
Assets/Scripts/Core/GameManager.cs     # Add crafting callback, time advancement
Assets/Scripts/Character/FeatDefinitions.cs # Remove IsPlaceholder from creation feats
Assets/Scripts/Character/CharacterStats.cs # Add XP spending method, crafting gold
Assets/Scripts/Core/SceneBootstrap.cs  # Initialize CraftableItemRegistry
```

3. All 8 Creation Feats — Complete Mechanics

3.1 Scribe Scroll

Property	Value
Prerequisite	Caster level 1st
Special	FREE for all spellcasters — granted automatically
Creates	Spell scrolls
Spell Level Limit	Any spell the caster knows/prepares
Base Price	Spell Level × Caster Level × 25 gp
Gold Cost	Base Price / 2
XP Cost	Base Price / 25
Time	1 day per 1,000 gp base price (minimum 1 day)
Spell Required	The spell being scribed (must know/prepare it)

Base Price Examples:

Scroll	Spell Lvl	CL	Base Price	Gold Cost	XP Cost	Time
Magic Missile (1st)	1	1	25 gp	12 gp	1 XP	1 day
Fireball (3rd)	3	5	375 gp	187 gp	15 XP	1 day
Stoneskin (4th)	4	7	700 gp	350 gp	28 XP	1 day
Heal (6th)	6	11	1,650 gp	825 gp	66 XP	2 days
Wish (9th)	9	17	3,825 gp	1,912 gp	153 XP	4 days

Note: 0th-level scrolls use spell level 0.5 for pricing: $0.5 \times CL \times 25 = CL \times 12.5$ gp.

3.2 Brew Potion

Property	Value
Prerequisite	Caster level 3rd
Creates	Potions
Spell Level Limit	3rd level or lower
Targeting	Must target one or more creatures (no area-only spells)
Base Price	Spell Level × Caster Level × 50 gp
Gold Cost	Base Price / 2
XP Cost	Base Price / 25
Time	1 day (always — potions are quick)
Spell Required	The spell being bottled

Base Price Examples:

Potion	Spell Lvl	CL	Base Price	Gold Cost	XP Cost
Cure Light Wounds	1	1	50 gp	25 gp	2 XP
Bull's Strength	2	3	300 gp	150 gp	12 XP
Fly	3	5	750 gp	375 gp	30 XP
Haste	3	5	750 gp	375 gp	30 XP

Note: 0th-level potions use spell level 0.5 for pricing: $0.5 \times CL \times 50 = CL \times 25$ gp .

Ineligible spells for potions:

- Spells with range “Personal” (unless also targets creatures)
- Spells with area-only effects (e.g., Fireball, Wall of Fire)
- Spells above 3rd level

3.3 Craft Wand

Property	Value
Prerequisite	Caster level 5th
Creates	Wands (50 charges)
Spell Level Limit	4th level or lower
Base Price	Spell Level × Caster Level × 750 gp
Gold Cost	Base Price / 2
XP Cost	Base Price / 25
Time	1 day per 1,000 gp base price (minimum 1 day)
Spell Required	The spell being stored
Charges	Always 50

Base Price Examples:

Wand	Spell Lvl	CL	Base Price	Gold Cost	XP Cost	Time
Cure Light Wounds	1	1	750 gp	375 gp	30 XP	1 day
Scorching Ray	2	3	4,500 gp	2,250 gp	180 XP	5 days
Fireball	3	5	11,250 gp	5,625 gp	450 XP	12 days
Stoneskin	4	7	21,000 gp	10,500 gp	840 XP	21 days

Note: 0th-level wands use spell level 0.5: $0.5 \times CL \times 750 = CL \times 375$ gp .

3.4 Craft Magic Arms and Armor

Property	Value
Prerequisite	Caster level 5th
Creates	Magic weapons, armor, shields
Requires	Masterwork base item (must already exist in inventory)
Base Price	Enhancement bonus squared × 2,000 gp (weapons) or × 1,000 gp (armor/shields)
Gold Cost	Base Price / 2
XP Cost	Base Price / 25
Time	1 day per 1,000 gp base price
Spells Required	Vary by enchantment (see table below)
CL Requirement	3 × enhancement bonus (minimum)

Enhancement Bonus Pricing:

Bonus	Weapon Base Price	Armor/ Shield Base Price	Gold Cost (W)	Gold Cost (A)	XP (W)	XP (A)	Time (W)	Time (A)
+1	2,000 gp	1,000 gp	1,000 gp	500 gp	80 XP	40 XP	2 days	1 day
+2	8,000 gp	4,000 gp	4,000 gp	2,000 gp	320 XP	160 XP	8 days	4 days
+3	18,000 gp	9,000 gp	9,000 gp	4,500 gp	720 XP	360 XP	18 days	9 days
+4	32,000 gp	16,000 gp	16,000 gp	8,000 gp	1,280 XP	640 XP	32 days	16 days
+5	50,000 gp	25,000 gp	25,000 gp	12,500 gp	2,000 XP	1,000 XP	50 days	25 days

Upgrading: When upgrading from +N to +M, the cost is the difference between the two prices (not the full +M cost).

Special Ability Pricing (adds equivalent bonus):

Weapon Ability	Equiv. Bonus	Required Spell
Flaming	+1	Flame Blade or Flame Strike or Fireball
Frost	+1	Chill Metal or Ice Storm
Shock	+1	Call Lightning or Lightning Bolt
Keen	+1	Keen Edge
Ghost Touch	+1	Plane Shift or Etherealness
Holy	+2	Holy Smite
Unholy	+2	Unholy Blight
Vorpal	+5	Circle of Death or Keen Edge

Armor Ability	Equiv. Bonus	Required Spell
Fortification, Light	+1	Limited Wish or Miracle
Shadow	+1	Invisibility or Shadow Walk
Silent Moves	+1	Silence
Spell Resistance (13)	+2	Spell Resistance
Fortification, Moderate	+3	Limited Wish or Miracle
Fortification, Heavy	+5	Limited Wish or Miracle

Maximum total bonus: +10 (enhancement + special abilities combined).

Required spell for base enhancement: +1 needs no specific spell; +2 + needs caster level $\geq 3 \times \text{bonus}$.

3.5 Craft Wondrous Item

Property	Value
Prerequisite	Caster level 3rd
Creates	Wondrous items (cloaks, boots, amulets, belts, etc.)
Base Price	Per item (from DMG table / existing database)
Gold Cost	Market Price / 2
XP Cost	Market Price / 25
Time	1 day per 1,000 gp market price
Spells Required	Vary by item (see DMG)

Common Wondrous Item Examples:

Item	Market Price	Gold Cost	XP Cost	Time	Required Spells
Cloak of Resistance +1	1,000 gp	500 gp	40 XP	1 day	Resistance
Boots of Speed	12,000 gp	6,000 gp	480 XP	12 days	Haste
Headband of Intellect +2	4,000 gp	2,000 gp	160 XP	4 days	Fox's Cunning
Belt of Giant Strength +4	16,000 gp	8,000 gp	640 XP	16 days	Bull's Strength
Amulet of Natural Armor +3	18,000 gp	9,000 gp	720 XP	18 days	Barkskin
Bag of Holding (Type I)	2,500 gp	1,250 gp	100 XP	3 days	Secret Chest
Carpet of Flying (5×5)	35,000 gp	17,500 gp	1,400 XP	35 days	Overland Flight
Iron Cobra	80,000 gp	40,000 gp	3,200 XP	80 days	Animate Objects, Geas/Quest

3.6 Craft Rod

Property	Value
Prerequisite	Caster level 9th
Creates	Rods
Base Price	Per rod (from DMG table / RodDatabase)
Gold Cost	Market Price / 2
XP Cost	Market Price / 25
Time	1 day per 1,000 gp market price
Spells Required	Vary by rod

Common Rod Examples:

Rod	Market Price	Gold Cost	XP Cost	Time	Required Spells
Metamagic (Lesser Extend)	3,000 gp	1,500 gp	120 XP	3 days	Extend Spell (feat)
Metamagic (Normal Empower)	32,500 gp	16,250 gp	1,300 XP	33 days	Empower Spell (feat)
Metamagic (Greater Quicken)	170,000 gp	85,000 gp	6,800 XP	170 days	Quicken Spell (feat)
Rod of Wonder	12,000 gp	6,000 gp	480 XP	12 days	Confusion, Fireball
Rod of Enemy Detection	23,500 gp	11,750 gp	940 XP	24 days	Detect Enemies

Metamagic Rod special rule: The crafter must possess the corresponding metamagic feat.

3.7 Craft Staff

Property	Value
Prerequisite	Caster level 12th
Creates	Staves (50 charges)
Base Price	Per staff (from DMG table / StaffDatabase)
Gold Cost	Market Price / 2
XP Cost	Market Price / 25
Time	1 day per 1,000 gp market price
Spells Required	ALL spells stored in the staff
Special	Crafter must supply all spells; CL must be $\geq$ highest spell CL

Staff Examples:

Staff	Market Price	Gold Cost	XP Cost	Time	Required Spells
Staff of Fire	17,750 gp	8,875 gp	710 XP	18 days	Burning Hands, Fireball, Wall of Fire
Staff of Healing	27,750 gp	13,875 gp	1,110 XP	28 days	Cure Serious, Lesser Restoration, Remove Disease
Staff of Power	211,000 gp	105,500 gp	8,440 XP	211 days	Magic Missile, Ray of Enfeeblement, Levitate, Lightning Bolt, Fireball, Cone of Cold, Hold Monster, Globe of Invulnerability, Wall of Force

3.8 Forge Ring

Property	Value
Prerequisite	Caster level 12th
Creates	Rings
Base Price	Per ring (from DMG table / RingDatabase)
Gold Cost	Market Price / 2
XP Cost	Market Price / 25
Time	1 day per 1,000 gp market price
Spells Required	Vary by ring

Ring Examples:

Ring	Market Price	Gold Cost	XP Cost	Time	Required Spells
Ring of Protection +1	2,000 gp	1,000 gp	80 XP	2 days	Shield of Faith
Ring of Protection +3	18,000 gp	9,000 gp	720 XP	18 days	Shield of Faith
Ring of Invisibility	20,000 gp	10,000 gp	800 XP	20 days	Invisibility
Ring of Spell Turning	98,280 gp	49,140 gp	3,931 XP	99 days	Spell Turning
Ring of Freedom of Movement	40,000 gp	20,000 gp	1,600 XP	40 days	Freedom of Movement

4. Cost System

4.1 Base Price Formulas

Creation Type	Base Price Formula
Scroll	Spell Level × Caster Level × 25 gp
Potion	Spell Level × Caster Level × 50 gp
Wand	Spell Level × Caster Level × 750 gp
Arms/Armor	Enhancement ² × 2,000 gp (weapon) or × 1,000 gp (armor)
Wondrous Item	Fixed per item (from database / DMG)
Rod	Fixed per item (from database / DMG)
Staff	Fixed per item (from database / DMG)
Ring	Fixed per item (from database / DMG)

4.2 Derived Costs

```
public static class CraftingCostCalculator
{
    /// Gold cost = market price / 2 (always)
    public static int GoldCost(int marketPrice) => Mathf.Max(1, marketPrice / 2);

    /// XP cost = market price / 25 (always)
    public static int XPCost(int marketPrice) => Mathf.Max(1, marketPrice / 25);

    /// Days = base price / 1000, minimum 1
    public static int DaysRequired(int basePrice) => Mathf.Max(1, basePrice / 1000);

    /// Scroll base price
    public static int ScrollBasePrice(int spellLevel, int casterLevel)
    {
        float effectiveLevel = spellLevel == 0 ? 0.5f : spellLevel;
        return Mathf.RoundToInt(effectiveLevel * casterLevel * 25f);
    }

    /// Potion base price
    public static int PotionBasePrice(int spellLevel, int casterLevel)
    {
        float effectiveLevel = spellLevel == 0 ? 0.5f : spellLevel;
        return Mathf.RoundToInt(effectiveLevel * casterLevel * 50f);
    }

    /// Wand base price (50 charges)
    public static int WandBasePrice(int spellLevel, int casterLevel)
    {
        float effectiveLevel = spellLevel == 0 ? 0.5f : spellLevel;
        return Mathf.RoundToInt(effectiveLevel * casterLevel * 750f);
    }

    /// Weapon enhancement base price
    public static int WeaponEnhancementBasePrice(int totalBonus)
        => totalBonus * totalBonus * 2000;

    /// Armor/shield enhancement base price
    public static int ArmorEnhancementBasePrice(int totalBonus)
        => totalBonus * totalBonus * 1000;

    /// Upgrade cost (difference between new and old)
    public static int UpgradeCost(int newBonus, int oldBonus, bool isWeapon)
    {
        int newPrice = isWeapon ? WeaponEnhancementBasePrice(newBonus) : ArmorEnhancementBasePrice(newBonus);
        int oldPrice = isWeapon ? WeaponEnhancementBasePrice(oldBonus) : ArmorEnhancementBasePrice(oldBonus);
        return newPrice - oldPrice;
    }
}
```

4.3 Material Component Costs

Some items have **additional material component costs** on top of the standard formula:

Item	Extra Material Cost	Reason
Scroll of Stoneskin	250 gp diamond dust	Spell's material component
Potion of Nondetection	50 gp	Spell's material component
Any item requiring costly component	Component cost	Added on top of base gold cost

Total Gold Cost = (Market Price / 2) + Material Component Costs

4.4 Maximum Crafting Per Day

Per DMG p.282: A character can craft items worth up to **1,000 gp per day**. Items costing more than 1,000 gp take multiple days. A character can only work on **one item at a time**.

5. Time System

5.1 Time Calculation

Days Required = $\text{ceil}(\text{Base Price} / 1,000)$ [minimum 1 day]

All potions take exactly 1 day regardless of price (DMG special rule).

5.2 Time Advancement Design

Crafting happens in **downtime mode** — when the player confirms crafting, time fast-forwards:

1. Calculate days required
2. Display confirmation: "This will take N days. Proceed?"
3. On confirm: advance game time by N days
4. Trigger appropriate resets for each day:
 - Daily ability resets (rings, wondrous items, rod uses)
 - Weekly resets every 7 days
 - Monthly resets every 30 days
5. Item appears in crafter's inventory

5.3 Time Examples

Item Being Crafted	Base Price	Days	Notes
Scroll of Fireball	375 gp	1	Below 1,000 → minimum 1 day
Potion of Fly	750 gp	1	Potions always 1 day
Wand of Magic Missile	750 gp	1	Below 1,000 → minimum 1 day
Ring of Protection +1	2,000 gp	2	
+1 Longsword	2,000 gp	2	
+3 Full Plate	9,000 gp	9	
+5 Greatsword	50,000 gp	50	Major time commitment
Staff of Power	211,000 gp	211	Nearly 7 months!

5.4 Interruption Rules

Per DMG: If crafting is interrupted, the crafter loses **materials and time spent so far** but not XP.

For the prototype: Crafting is atomic (instant fast-forward), so interruption doesn't apply in the current design. If we add multi-encounter interruptions later:

```
public class CraftingProject
{
    public int DaysCompleted;
    public int DaysTotal;
    public int GoldSpentSoFar; // Lost on interruption
    public int XPReserved;    // Refunded on interruption
    public bool IsComplete => DaysCompleted >= DaysTotal;
}
```

6. Spell Prerequisite System

6.1 How Prerequisites Work

Each magic item has **required spells** that the crafter must provide. The crafter can supply spells from:

1. **Known/Prepared Spells** — Check via `SpellcastingComponent.GetKnownSpellsForClass()`
2. **Class Spell List Access** — Classes that “know all” (Cleric, Druid) via `SpellcastingComponent.ClassKnowsAllSpells()`
3. **Spell Substitution** — Missing spells increase Spellcraft DC by +5 per missing spell (DMG p.215)

4. **Another Character** — A different party member can supply a missing spell (they must be present)

6.2 Spell Substitution Rule (DMG p.215)

“If the creator doesn’t have a required spell (or doesn’t know it), he can attempt to complete the creation anyway by increasing the DC of the Spellcraft check by +5 for each missing spell.”

Implementation:

```
public static class CraftingValidator
{
    /// Check spell prerequisites. Returns (metSpells, missingSpells, substitutionDC)
    public static SpellPrereqResult CheckSpellPrereqs(
        CharacterController crafter,
        List<string> requiredSpellIds,
        List<CharacterController> partyMembers)
    {
        var met = new List<string>();
        var missing = new List<string>();

        foreach (string spellId in requiredSpellIds)
        {
            bool found = CrafterKnowsSpell(crafter, spellId);
            if (!found)
                found = PartyMemberKnowsSpell(partyMembers, spellId, crafter);

            if (found) met.Add(spellId);
            else missing.Add(spellId);
        }

        int substitutionDC = 5 * missing.Count;
        return new SpellPrereqResult(met, missing, substitutionDC);
    }
}
```

6.3 Per-Category Spell Requirements

Scrolls, Potions, Wands

- Require **exactly the spell being stored** — no substitution allowed for the primary spell
- The crafter must know/prepare the specific spell

Magic Arms and Armor

- +1 enhancement: No specific spell, just Craft Magic Arms and Armor feat
- Special abilities (Flaming, Frost, etc.): Require specific spells (see table in §3.4)
- Multiple abilities require all prerequisite spells

Wondrous Items, Rods, Rings, Staves

- Each item has a specific list of prerequisite spells defined in the DMG
- These will be stored in `CraftableItemDefinition.RequiredSpellIds`

6.4 Spell Prerequisite Data Storage

Add to the crafting data:

```

public class CraftableItemDefinition
{
    public string ItemId;           // Links to ItemDatabase
    public CraftingFeatType RequiredFeat;
    public int MinimumCasterLevel;
    public List<string> RequiredSpellIds; // Spell IDs from SpellDatabase
    public List<string> RequiredFeatNames; // Additional feat prerequisites
    public int MaterialComponentCostGp; // Extra material costs
    public bool RequiresMasterworkBase; // For arms/armor
    public string MasterworkBaseItemId; // Which base item to consume
}

```

7. Validation System

7.1 Complete Validation Checklist

Before crafting can begin, the system validates ALL of:

VALIDATION PIPELINE
1. Has the required Item Creation feat? → CharacterStats.HasFeat("Brew Potion")
2. Meets minimum caster level? → CL ≥ item's required CL → Feat CL: Scroll≥1, Potion≥3, Wand/Arms≥5, Rod≥9, Staff/Ring≥12
3. Has required spells? → Direct: crafter knows the spell → Party: another member supplies it → Substitution: +5 DC per missing
4. Has required additional feats? → Metamagic rods need metamagic feat → Some items need specific feats
5. Has sufficient gold? → Gold ≥ (MarketPrice / 2) + materials
6. Has sufficient XP? → XP ≥ (MarketPrice / 25) → XP after spending ≥ level threshold
7. Has masterwork base? (Arms/Armor only) → Item in inventory with Masterwork=true
8. Doesn't violate maximum limits? → Max +10 total bonus (Arms/Armor) → Max 4th level spell (Wands) → Max 3rd level spell (Potions) → Potion must target creatures

7.2 Validation Result Structure

```
public class CraftingValidationResult
{
    public bool IsValid;
    public bool HasFeat;
    public bool MeetsCasterLevel;
    public int RequiredCasterLevel;
    public int ActualCasterLevel;
    public bool HasAllSpells;
    public List<string> MetSpells;
    public List<string> MissingSpells;
    public int SpellSubstitutionDC;
    public bool HasRequiredFeats;
    public List<string> MissingFeats;
    public bool HasGold;
    public int GoldRequired;
    public int GoldAvailable;
    public bool HasXP;
    public int XPRequired;
    public int XPAvailable;
    public int XPMinimumForLevel; // Can't go below this
    public bool HasMasterworkBase; // Arms/Armor only
    public string FailureReason; // Human-readable summary
}
```

8. UI / Menu Design

8.1 Pre-Combat Hub Integration

Modify `PreCombatHubUI` to add a new button:

Current Buttons:

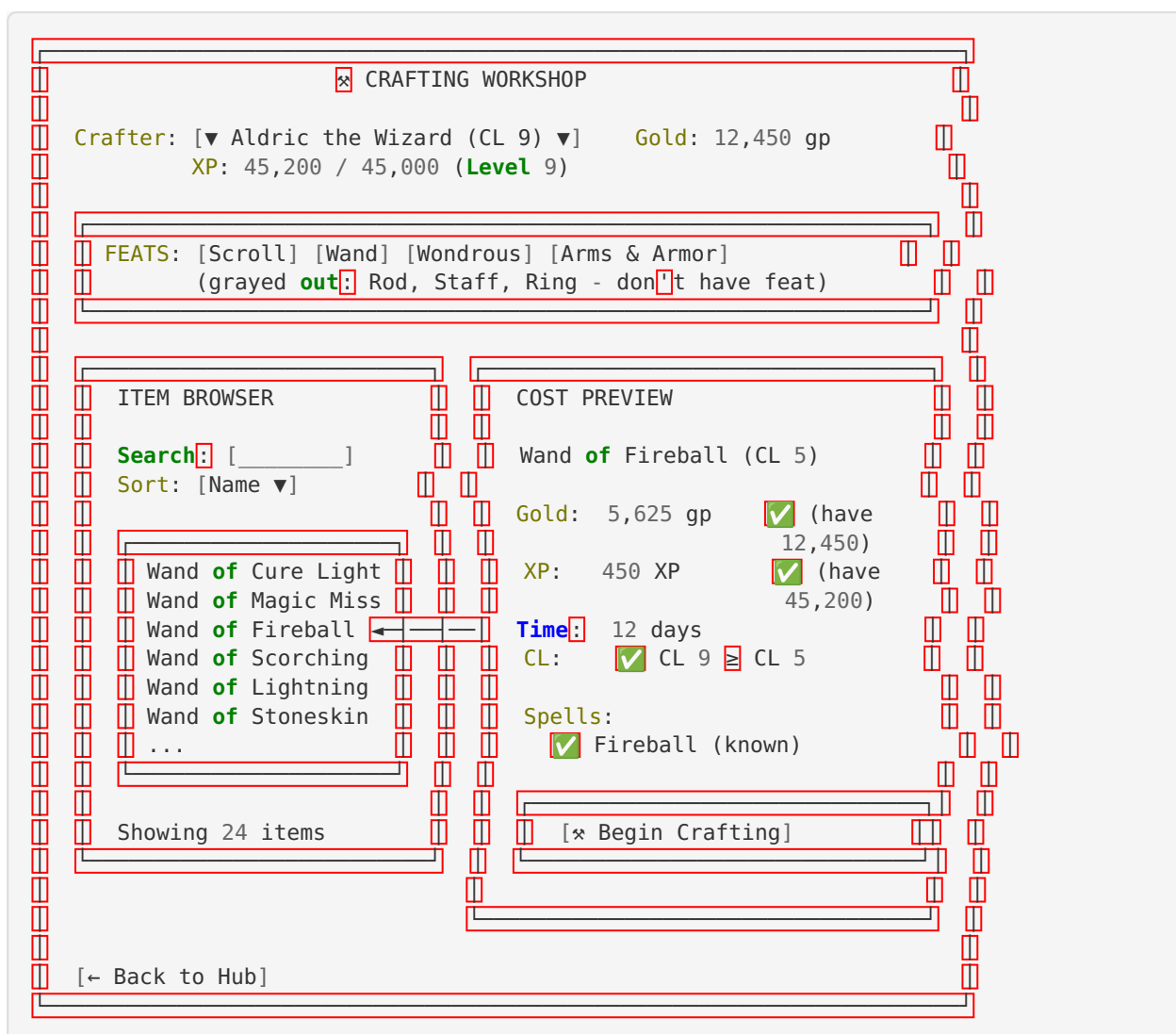
-  Manage Inventory (Stash)
-  Open Store
-  Prepare Spells
- × Start Encounter
- ← Back to Encounter Selection

New Layout (add between Prepare Spells and Start Encounter):

-  Manage Inventory (Stash)
-  Open Store
-  Prepare Spells
- ✂ Crafting Workshop ← NEW
- × Start Encounter
- ← Back to Encounter Selection

The button is **always visible** but shows “(No crafters)” and is grayed out if no party member has any item creation feat.

8.2 Crafting Workshop — Main Layout



8.3 Item Browser Panel

Features:

- **Search field** — Filter items by name substring
- **Sort** — By name (A-Z), price (low-high, high-low), spell level, caster level
- **Category filter** — Subcategories per feat (e.g., for Wondrous: head, neck, back, etc.)
- **Availability indicator** — Items the character can't craft show with ⚠ icon and red tint
- **Scroll list** — Standard Unity scroll view, items as clickable rows
- **Selected item** — Highlighted row, details shown in Cost Preview

Each row displays:

[Icon] Item Name Gold: X gp | XP: X | Time: X days

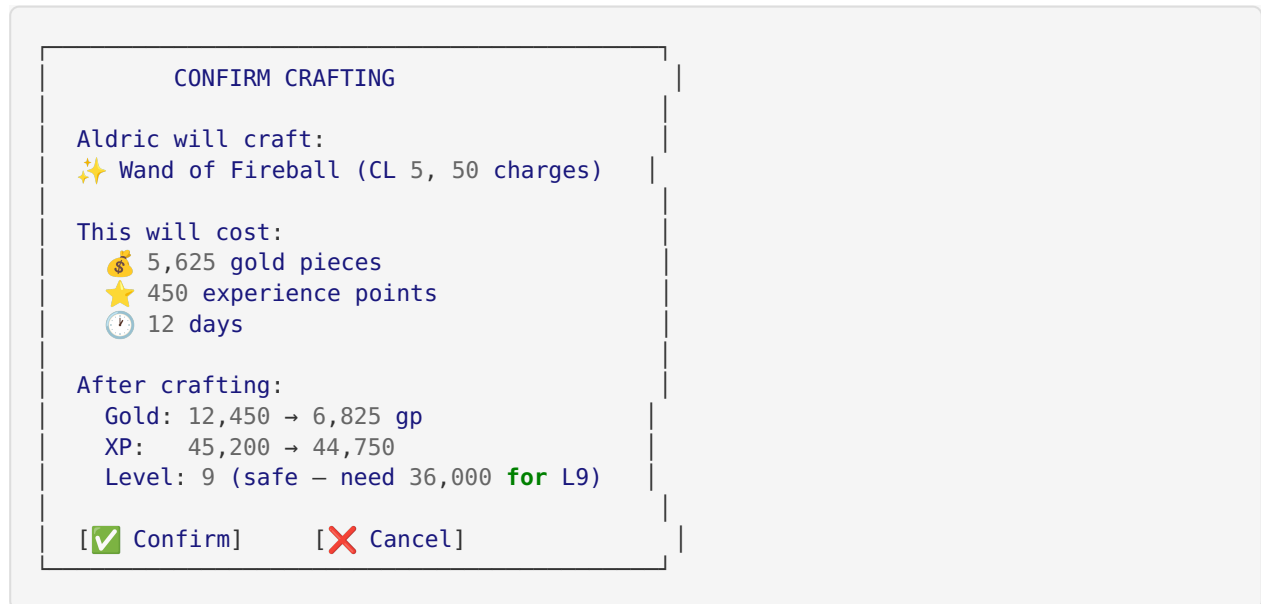
8.4 Cost Preview Panel

Shows full details for the selected item:

- Item name and description (truncated)
- **Gold cost** — amount + [✓/✗] status (with "have X gp" note)
- **XP cost** — amount + [✓/✗] status (with "have X XP" note)

- **XP safety** — Warning if XP would drop below level threshold
- **Time** — Days required
- **Caster Level** — Required vs. actual, / 
- **Spell prerequisites** — Each spell listed with  known /  party member /  missing (+5 DC)
- **Additional feats** — If required (metamagic for rods)
- **Masterwork base** — For arms/armor, shows if base item is in inventory
- **“Begin Crafting” button** — Enabled only when all requirements met (or substitution accepted)

8.5 Confirmation Dialog



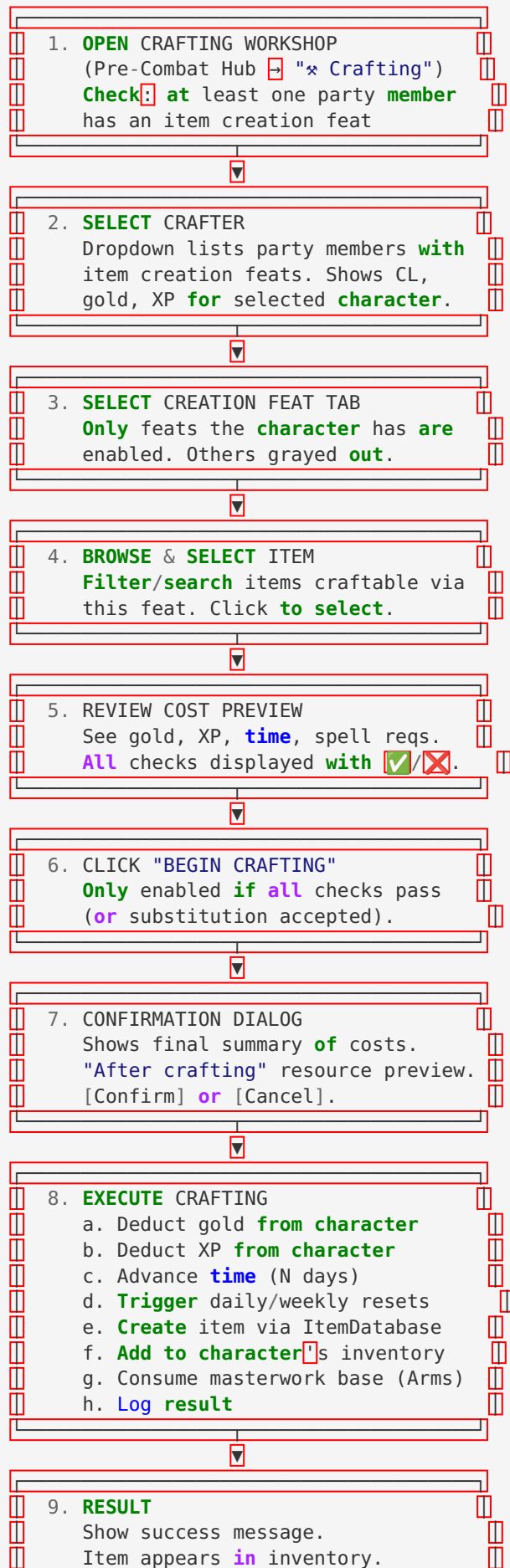
8.6 Crafting In Progress (Fast-Forward) Animation

When crafting executes:

1. Brief overlay: “⚡ Crafting in progress... (12 days)”
 2. Progress bar fills (cosmetic — actually instant)
 3. Result popup: “ Aldric crafted Wand of Fireball! (50 charges)”
 4. Item added to inventory
 5. Return to Crafting Workshop (can craft more)
-

9. Crafting Workflow

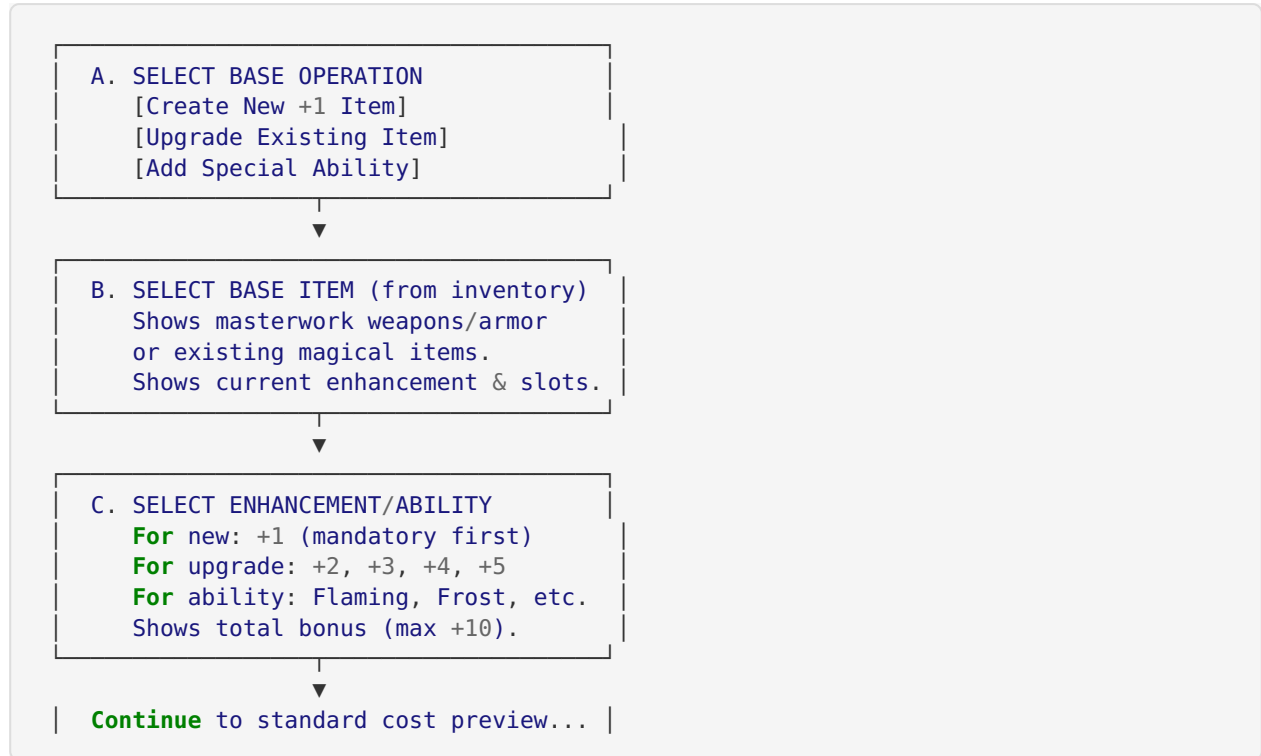
9.1 Complete Step-by-Step Flow



Return to Workshop (craft more).

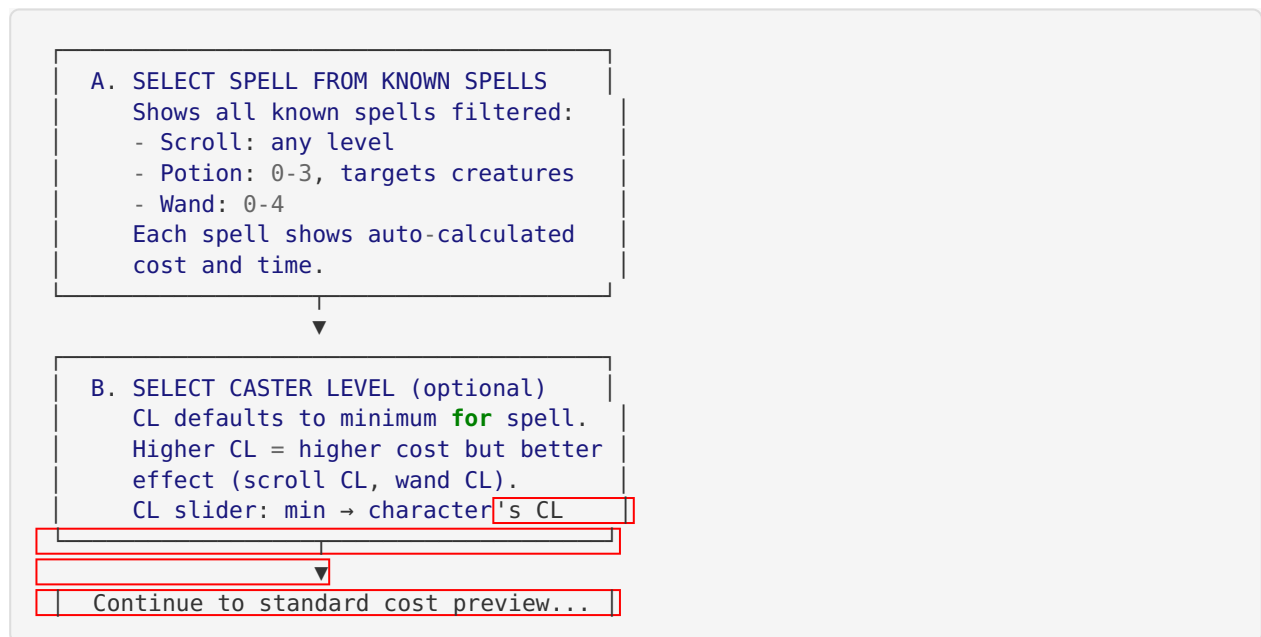
9.2 Special Workflow: Craft Magic Arms and Armor

This feat has a unique sub-workflow because it operates on existing items:



9.3 Special Workflow: Scribe Scroll / Brew Potion / Craft Wand

These feats create items from **the caster's own spells**, not from a database list:



10. Item Filtering — What Can Be Crafted

10.1 Per-Feat Item Sources

Feat	Source Database	Filter Logic
Scribe Scroll	Character's known spells	All spell levels; generate scroll ItemData on-the-fly
Brew Potion	Character's known spells	Level $\leq$ 3, must target creatures (not area-only, not personal)
Craft Wand	Character's known spells	Level $\leq$ 4
Craft Magic Arms & Armor	Character's inventory (masterwork base) + EnchantmentFactory	Masterwork items for new; existing magic items for upgrade
Craft Wondrous Item	<code>WondrousItemDataBase.GetAllItems()</code>	All wondrous items with <code>CrafterCasterLevel $\leq$ character CL</code>
Craft Rod	<code>RodDatabase.GetAllRods()</code>	All rods; metamagic rods also need the metamagic feat
Craft Staff	<code>StaffDataBase.GetAllStaves()</code>	All staves; crafter must know ALL spells in staff
Forge Ring	<code>RingDatabase.GetAllRings()</code>	All rings with <code>CrafterCasterLevel $\leq$ character CL</code>

10.2 Filtering Implementation

```
public static class CraftableItemRegistry
{
    /// Get all items a character can craft with a specific feat
    public static List<CraftableItemDefinition> GetCraftableItems(
        CharacterController crafter,
        CraftingFeatType feat)
    {
        switch (feat)
        {
            case CraftingFeatType.ScribeScroll:
                return GetScrollCraftables(crafter);
            case CraftingFeatType.BrewPotion:
                return GetPotionCraftables(crafter);
            case CraftingFeatType.CraftWand:
                return GetWandCraftables(crafter);
            case CraftingFeatType.CraftMagicArmsAndArmor:
                return GetArmsArmorCraftables(crafter);
            case CraftingFeatType.CraftWondrousItem:
                return GetWondrousCraftables(crafter);
            case CraftingFeatType.CraftRod:
                return GetRodCraftables(crafter);
            case CraftingFeatType.CraftStaff:
                return GetStaffCraftables(crafter);
            case CraftingFeatType.ForgeRing:
                return GetRingCraftables(crafter);
            default:
                return new List<CraftableItemDefinition>();
        }
    }
}
```

10.3 Dynamic Item Generation (Scrolls, Potions, Wands)

These three feats generate items **dynamically from the character's spell list**, not from a fixed database:

```

private static List<CraftableItemDefinition> GetScrollCraftables(CharacterController
crafter)
{
    var result = new List<CraftableItemDefinition>();
    var sc = crafter.GetComponent<SpellcastingComponent>();
    if (sc == null) return result;

    foreach (var spell in sc.GetAllKnownSpells().Select(SpellDatabase.GetSpell).Where(
s => s != null))
    {
        int cl = GetMinimumCasterLevel(spell);
        result.Add(new CraftableItemDefinition
        {
            ItemId = $"scroll_{spell.SpellId}",
            DisplayName = $"Scroll of {spell.Name}",
            RequiredFeat = CraftingFeatType.ScribeScroll,
            MinimumCasterLevel = Mathf.Max(1, cl),
            MarketPrice = CraftingCostCalculator.ScrollBasePrice(spell.SpellLevel, cl)
        },

        RequiredSpellIds = new List<string> { spell.SpellId },
        IsDynamic = true,
        SourceSpellId = spell.SpellId,
        SourceSpellLevel = spell.SpellLevel,
        DefaultCasterLevel = cl
        ));
    }
    return result;
}

```

11. Special Requirements

11.1 Masterwork Base Item (Arms & Armor)

D&D 3.5e Rule: You can only enchant a **masterwork** weapon, armor, or shield. The masterwork item is consumed/transformed during crafting.

Implementation:

1. When “Craft Magic Arms and Armor” tab is selected, scan the crafter’s inventory for masterwork items
2. Display available base items
3. On crafting, remove the masterwork base item and replace with the magical version
4. The masterwork cost (typically 300 gp for weapons, 150 gp for armor) is NOT included in enchantment cost

```

public static List<ItemData> GetMasterworkBaseItems(Inventory inventory)
{
    return inventory.GetAllItems()
        .Where(item => item.IsMasterwork && (item.IsWeapon || item.IsArmor || item.IsShield))
        .Where(item => item.EnhancementBonus == 0) // Not already magical
        .ToList();
}

// For upgrading existing magical items:
public static List<ItemData> GetUpgradeableItems(Inventory inventory)
{
    return inventory.GetAllItems()
        .Where(item => item.IsWeapon || item.IsArmor || item.IsShield)
        .Where(item => item.EnhancementBonus >= 1 && item.EnhancementBonus < 5)
        .ToList();
}

```

11.2 Enhancement Progression Rules

Magic weapons and armor must follow a strict progression:

1. **First enhancement** must be **+1** (can't jump to +2)
2. Enhancement bonus: +1 to +5
3. Special abilities add equivalent bonus (+1 for Flaming, +2 for Holy, etc.)
4. **Total equivalent bonus** (enhancement + special abilities) cannot exceed **+10**
5. An item must have at least **+1** enhancement before adding special abilities

11.3 Metamagic Rod Special Rule

Crafting a metamagic rod requires the crafter to **possess the corresponding metamagic feat**:

```

// Validation for metamagic rods
if (rod.RodIsMetamagic)
{
    string metamagicFeatName = MetamagicData.GetDisplayName(rod.RodMetamagicType);
    if (!crafter.Stats.HasFeat(metamagicFeatName))
    {
        result.HasRequiredFeats = false;
        result.MissingFeats.Add(metamagicFeatName);
    }
}

```

11.4 Staff Special Rule

The crafter must be able to **cast ALL spells** stored in the staff:

```

public static bool CanCraftStaff(CharacterController crafter, StaffDefinition staff)
{
    var sc = crafter.GetComponent<SpellcastingComponent>();
    foreach (var spellEntry in staff.Spells)
    {
        if (!sc.KnowsSpell(SpellDatabase.GetSpell(spellEntry.SpellId), crafter.Stats.PrimaryClass))
            return false;
    }
    return true;
}

```

11.5 Potion Targeting Restriction

Potions can only contain spells that **target one or more creatures** (not area-only, not personal range):

```

public static bool IsValidPotionSpell(SpellData spell)
{
    if (spell.SpellLevel > 3) return false;
    if (spell.Range == SpellRange.Personal) return false; // Exception: some Personal spells are allowed
    if (spell.TargetType == SpellTargetType.AreaOnly) return false;
    return true;
}

```

12. XP Management

12.1 Preventing Level Loss

Critical Rule: Spending XP on item creation must NEVER cause a character to drop below their current level's XP threshold.

```

D&D 3.5e XP Table:
Level 1: 0 XP
Level 2: 1,000 XP
Level 3: 3,000 XP
Level 4: 6,000 XP
Level 5: 10,000 XP
Level 6: 15,000 XP
Level 7: 21,000 XP
Level 8: 28,000 XP
Level 9: 36,000 XP
Level 10: 45,000 XP
...
Formula: Level N requires N × (N-1) × 500 XP

```

Implementation using existing `ExperienceCalculator` :

```

public static bool CanSpendXP(CharacterStats stats, int xpCost)
{
    int currentLevel = stats.Level;
    int minXPForLevel = ExperienceCalculator.GetXPForLevel(currentLevel);
    int xpAfterSpending = stats.ExperiencePoints - xpCost;
    return xpAfterSpending >= minXPForLevel;
}

public static int MaxSpendableXP(CharacterStats stats)
{
    int currentLevel = stats.Level;
    int minXPForLevel = ExperienceCalculator.GetXPForLevel(currentLevel);
    return stats.ExperiencePoints - minXPForLevel;
}

```

12.2 XP Spending Method (Add to CharacterStats)

```

/// <summary>
/// Spend XP on item creation. Returns false if would cause level loss.
/// </summary>
public bool SpendXP(int amount)
{
    if (amount <= 0) return false;
    int minXP = ExperienceCalculator.GetXPForLevel(Level);
    if (ExperiencePoints - amount < minXP)
    {
        Debug.LogWarning($"[XP] Cannot spend {amount} XP – would drop below level {Level} threshold ({minXP})");
        return false;
    }
    ExperiencePoints -= amount;
    Debug.Log($"[XP] {CharacterName} spent {amount} XP on crafting. Remaining: {ExperiencePoints}");
    return true;
}

```

12.3 UI Safety Warnings

The Cost Preview panel shows XP warnings:

XP Situation	Display
XP plenty	✓ 450 XP (have 45,200; floor: 36,000)
XP tight	⚠ 450 XP (have 36,500; floor: 36,000 – only 500 available!)
XP insufficient	✗ 450 XP (have 36,300; floor: 36,000 – only 300 available!)
Would cause level loss	⊘ 450 XP (WOULD LOSE LEVEL 9! Need 36,000 minimum)

13. Time Management

13.1 Days Passing

When crafting completes, the game advances N days. Each day triggers:

```
public static void AdvanceCraftingDays(int days, List<CharacterController> party)
{
    for (int d = 0; d < days; d++)
    {
        _gameDayCount++;

        // Daily resets (same as rest)
        RingActivationManager.OnRest(party);
        WondrousItemActivation.OnRest(party);
        RodDatabase.ResetDailyUses();

        // Weekly resets every 7 days
        if (_gameDayCount % 7 == 0)
        {
            RingUseTracker.Instance?.OnWeeklyReset();
            WondrousItemActivation.OnWeeklyReset(party);
            RodDatabase.ResetWeeklyUses();
        }

        // Monthly resets every 30 days
        if (_gameDayCount % 30 == 0)
        {
            WondrousItemActivation.OnMonthlyReset(party);
        }

        // Heal naturally (1 HP per level per day of rest)
        foreach (var pc in party)
        {
            if (pc.Stats != null)
                pc.Stats.HealDamage(pc.Stats.Level);
        }
    }
}
```

13.2 Day Counter

Add to `GameManager` :

```
private int _gameDayCount = 0;
public int GameDayCount => _gameDayCount;

/// Advance game days (called from CraftingExecutor and rest)
public void AdvanceDays(int count)
{
    CraftingTimeCalculator.AdvanceCraftingDays(count, GetAllPCs());
}
```

13.3 UI Time Display

Show the player clear information about time cost:

Time: 12 days

- └ Items reset: daily abilities, rod uses, ring charges
 - └ Natural healing: $12 \times \text{Level HP recovered}$
 - └ **If** > 7 days: weekly resets also trigger
-

14. Integration Points

14.1 Existing System Connections

System	Integration	How
ItemDatabase	Retrieve item templates for crafting	<code>ItemDatabase.Get(id)</code> , <code>ItemDatabase.CloneItem(id)</code>
RodDatabase	List all craftable rods	<code>RodDatabase.GetAllRods()</code> , market prices from <code>ItemData.PriceGp</code>
RingDatabase	List all craftable rings	<code>RingDatabase.GetAllRings()</code>
WondrousItemDatabase	List all craftable wondrous items	<code>WondrousItemDatabase.GetAllItems()</code>
StaffDatabase	List all craftable staves	<code>StaffDatabase.GetAllStaves()</code> , spell lists from <code>StaffDefinition</code>
SpecificItemDatabase	Excluded — specific items cannot be player-crafted	N/A
SpellDatabase	Spell lookups for prerequisites	<code>SpellDatabase.GetSpell(id)</code>
SpellcastingComponent	Check known spells for pre-requisites	<code>GetKnownSpellsForClass()</code> , <code>GetAllKnownSpells()</code>
CharacterStats	Feat checks, XP, gold, level, caster level	<code>HasFeat()</code> , <code>ExperiencePoints</code> , <code>ComponentGold</code> , <code>Level</code>
FeatDefinitions	Validate item creation feat ownership	Remove <code>IsPlaceholder</code> flag, feat already defined
Inventory	Add crafted items, check masterwork bases	<code>AddItem()</code> , <code>GetAllItems()</code>
PreCombatHubUI	Add Crafting Workshop button	New callback: <code>_onOpenCraftingWorkshop</code>
GameManager	Time advancement, day counter	New <code>AdvanceDays()</code> method
ExperienceCalculator	XP floor calculations	<code>GetXpForLevel()</code>
EnchantmentFactory	Create enchantments for crafted weapons/armor	<code>CreateEnchantment()</code> , <code>ApplyEnchantment()</code>
SceneBootstrap	Initialize CraftableItemRegistry	<code>CraftableItemRegistry.Init()</code> call

14.2 Gold System

Currently `CharacterStats.ComponentGold` is used for spell components. For crafting:

Option A (Recommended): Use the same `ComponentGold` pool — crafting materials come from the same gold reserve as spell components.

Option B: Add a separate `CraftingGold` field — more accurate but adds complexity.

Decision: Option A. Rename in UI to “Gold” and document that it covers both spell components and crafting materials. Add `CraftingGoldSpent` tracking field for statistics.

```
// Add to CharacterStats
public int TotalCraftingGoldSpent = 0;
public int TotalCraftingXPSpent = 0;
public int TotalItemsCrafted = 0;
```

14.3 Crafted Item Identification

Crafted items should be distinguishable from found items:

```
// Add to ItemData
public bool IsCraftedByPlayer = false;
public string CraftedByCharacterName = "";
public int CraftedOnDay = 0;
```

This allows the tooltip to show “Crafted by Aldric (Day 23)” and enables future features like selling crafted items.

15. Data Structures

15.1 CraftingFeatType Enum

```
public enum CraftingFeatType
{
    ScribeScroll,
    BrewPotion,
    CraftWand,
    CraftMagicArmsAndArmor,
    CraftWondrousItem,
    CraftRod,
    CraftStaff,
    ForgeRing
}
```

15.2 CraftableItemDefinition

```

/// <summary>
/// Template for a craftable item. Pre-built for database items,
/// dynamically generated for scrolls/potions/wands.
/// </summary>
public class CraftableItemDefinition
{
    // Identity
    public string ItemId;           // Links to ItemDatabase or generated ID
    public string DisplayName;      // Human-readable name
    public string Description;      // Short description
    public CraftingFeatType RequiredFeat;

    // Costs
    public int MarketPrice;         // Full market price
    public int GoldCost => CraftingCostCalculator.GoldCost(MarketPrice);
    public int XPCost => CraftingCostCalculator.XPCost(MarketPrice);
    public int DaysRequired => RequiredFeat == CraftingFeatType.BrewPotion
        ? 1 : CraftingCostCalculator.DaysRequired(MarketPrice);
    public int MaterialComponentCostGp; // Additional material costs

    // Prerequisites
    public int MinimumCasterLevel;
    public List<string> RequiredSpellIds = new List<string>();
    public List<string> RequiredFeatNames = new List<string>();

    // Arms & Armor specific
    public bool RequiresMasterworkBase;
    public string MasterworkBaseItemId;
    public int EnhancementBonus;    // For weapons/armor
    public string SpecialAbilityId; // For adding specific enchantments

    // Dynamic items (scrolls/potions/wands)
    public bool IsDynamic;
    public string SourceSpellId;
    public int SourceSpellLevel;
    public int DefaultCasterLevel;
    public int CustomCasterLevel; // Player-selected CL (scroll/wand)

    // Category (for UI filtering)
    public string Category;         // "Head", "Neck", "Weapon", etc.
    public string SubCategory;      // More specific grouping
}

```

15.3 CraftingProject

```

/// <summary>
/// Represents an in-progress or completed crafting project.
/// Used for execution and history tracking.
/// </summary>
public class CraftingProject
{
    // Identity
    public string ProjectId = System.Guid.NewGuid().ToString();
    public CraftableItemDefinition Definition;
    public CharacterController Crafter;

    // State
    public CraftingProjectState State = CraftingProjectState.NotStarted;
    public int DaysCompleted;
    public int DaysTotal;

    // Costs committed
    public int GoldCost;
    public int XPCost;
    public int MaterialCost;

    // Validation snapshot
    public CraftingValidationResult ValidationResult;

    // Result
    public ItemData CraftedItem;           // Set on completion
    public int CompletionDay;             // Game day when completed

    // Arms/Armor specific
    public ItemData ConsumedBaseItem;      // Masterwork base consumed
    public int PreviousEnhancement;        // For upgrade tracking
}

public enum CraftingProjectState
{
    NotStarted,
    InProgress,    // For future multi-session crafting
    Completed,
    Failed,
    Cancelled
}

```

15.4 Crafting History (Optional)

```

/// <summary>
/// Tracks all completed crafting projects for statistics/achievements.
/// </summary>
public static class CraftingHistory
{
    private static List<CraftingProject> _completedProjects = new
List<CraftingProject>();

    public static void RecordCompletion(CraftingProject project)
    {
        _completedProjects.Add(project);
        Debug.Log($"[Crafting] History: {_completedProjects.Count} items crafted
total");
    }

    public static int TotalItemsCrafted => _completedProjects.Count;
    public static int TotalGoldSpent => _completedProjects.Sum(p => p.GoldCost);
    public static int TotalXPSpent => _completedProjects.Sum(p => p.XPCost);
    public static int TotalDaysSpent => _completedProjects.Sum(p => p.DaysTotal);
}

```

16. Implementation Phases

Phase 1: Core Infrastructure (3-4 days)

Files: CraftingFeatType.cs , CraftableItemDefinition.cs , CraftingCostCalculator.cs , CraftingTimeCalculator.cs , CraftingValidator.cs

Tasks:

- [] Create CraftingFeatType enum
- [] Create CraftableItemDefinition data class
- [] Implement CraftingCostCalculator with all formulas (scroll, potion, wand, arms, fixed-price)
- [] Implement CraftingTimeCalculator with day calculations
- [] Implement CraftingValidator with full validation pipeline
- [] Add SpendXP() method to CharacterStats
- [] Add crafting tracking fields to CharacterStats
- [] Write unit tests for cost calculations
- [] Write unit tests for validation logic

Acceptance: All cost/time calculations match DMG tables; validation catches all invalid states.

Phase 2: Feat Activation (1-2 days)

Files: FeatDefinitions.cs , FeatManager.cs

Tasks:

- [] Remove IsPlaceholder = true from all 8 item creation feats
- [] Add GetItemCreationFeats(CharacterStats) to FeatManager
- [] Add GetCasterLevel(CharacterStats) helper (handles Ranger/Paladin half-CL)
- [] Implement Scribe Scroll auto-grant for all spellcasters
- [] Verify feat prerequisites (CasterLevel type) work correctly
- [] Write tests for feat validation

Acceptance: Feats show in character sheet; prerequisites correctly evaluated.

Phase 3: CraftableItemRegistry — Database Items (3-4 days)

Files: `CraftableItemRegistry.cs` , `CraftableItemDefinition.cs`

Tasks:

- [] Build registry initialization in `CraftableItemRegistry.Init()`
- [] Register all wondrous items from `WondrousItemDatabase` with spell prerequisites
- [] Register all rods from `RodDatabase` with spell prerequisites
- [] Register all rings from `RingDatabase` with spell prerequisites
- [] Register all staves from `StaffDatabase` with spell prerequisites
- [] Define spell prerequisites for top 50 most common items
- [] Define spell prerequisites for remaining items (can use “unknown” fallback)
- [] Wire initialization into `SceneBootstrap.cs`

Acceptance: Registry returns correct craftable items for each feat type.

Phase 4: CraftableItemRegistry — Dynamic Items (2-3 days)

Files: `CraftableItemRegistry.cs`

Tasks:

- [] Implement `GetScrollCraftables()` from character known spells
- [] Implement `GetPotionCraftables()` with targeting filter
- [] Implement `GetWandCraftables()` with level 4 cap
- [] Implement scroll/potion/wand item creation (generate `ItemData` on-the-fly)
- [] Handle 0th-level spell pricing (×0.5)
- [] Handle caster level selection for scrolls/wands
- [] Write tests for dynamic item generation

Acceptance: Scrolls/potions/wands correctly generated from known spells with proper pricing.

Phase 5: CraftingExecutor (2-3 days)

Files: `CraftingExecutor.cs` , `CraftingProject.cs` , `GameManager.cs`

Tasks:

- [] Implement `CraftingExecutor.Execute(CraftingProject)` — main execution method
- [] Implement gold deduction (from `ComponentGold`)
- [] Implement XP deduction (with level-loss prevention)
- [] Implement time advancement (call `GameManager.AdvanceDays()`)
- [] Implement daily/weekly/monthly reset triggers during time advancement
- [] Implement item creation and inventory addition
- [] Implement masterwork base item consumption (arms/armor)
- [] Implement item enchantment creation (arms/armor upgrades)
- [] Add crafted-item metadata (`IsCraftedByPlayer` , etc.)
- [] Implement `CraftingHistory` tracking
- [] Add `AdvanceDays()` to `GameManager`
- [] Wire rest/reset systems into time advancement
- [] Write integration tests for full crafting flow

Acceptance: Complete crafting flow works end-to-end — deducts costs, advances time, creates item.

Phase 6: Crafting Workshop UI — Main Panel (3-4 days)

Files: `CraftingWorkshopUI.cs` , `PreCombatHubUI.cs`

Tasks:

- [] Add “✂ Crafting Workshop” button to `PreCombatHubUI`
- [] Add `_onOpenCraftingWorkshop` callback to `PreCombatHubUI.Open()`
- [] Create `CraftingWorkshopUI` fullscreen panel (matching existing UI style)
- [] Implement character selector dropdown (party members with creation feats)
- [] Implement feat tab bar (only enabled feats)
- [] Implement character info display (CL, gold, XP, level)
- [] Implement “← Back to Hub” navigation
- [] Wire `GameManager` to open/close `CraftingWorkshopUI`

Acceptance: UI opens from Pre-Combat Hub; character and feat selection works.

Phase 7: Item Browser Panel (2-3 days)

Files: `CraftingItemBrowserPanel.cs`

Tasks:

- [] Implement scrollable item list from `CraftableItemRegistry`
- [] Implement search/filter text field
- [] Implement sort options (name, price, level)
- [] Implement category filters (subcategories per feat type)
- [] Implement item row display (icon, name, cost summary)
- [] Implement availability indicators (✓/⚠/✗)
- [] Implement item selection (click to select, highlight)
- [] Connect selection to Cost Preview panel

Acceptance: Items display correctly per feat; filtering/sorting works; selection triggers preview.

Phase 8: Cost Preview & Confirmation (2-3 days)

Files: `CraftingCostPreviewPanel.cs` , `CraftingConfirmationDialog.cs`

Tasks:

- [] Implement cost preview panel showing all validation details
- [] Display gold cost with availability check
- [] Display XP cost with level-loss warning
- [] Display time (days) with reset information
- [] Display caster level check
- [] Display spell prerequisites (per-spell ✓/⚠/✗)
- [] Display additional feat requirements
- [] Display masterwork base item (arms/armor)
- [] Implement “Begin Crafting” button (enabled only when valid)
- [] Implement confirmation dialog with final cost summary
- [] Implement crafting progress animation (brief overlay)
- [] Implement result notification

Acceptance: Full cost preview displays correctly; confirmation flow works; crafting executes.

Phase 9: Arms & Armor Special UI (2-3 days)

Files: `CraftingWorkshopUI.cs` (extend), `CraftingItemBrowserPanel.cs` (extend)

Tasks:

- [] Implement “Create New / Upgrade / Add Ability” sub-menu
- [] Implement masterwork base item selector (from inventory)
- [] Implement enhancement level selector (+1 through +5)
- [] Implement special ability selector with prerequisite display
- [] Implement total bonus display (max +10 rule)
- [] Implement upgrade cost calculation (difference pricing)
- [] Implement base item consumption confirmation
- [] Write tests for arms/armor crafting flow

Acceptance: Full arms/armor crafting workflow works including upgrades.

Phase 10: Polish, Testing, & Documentation (2-3 days)**Tasks:**

- [] Comprehensive play-testing of all 8 feat types
- [] Edge case testing (0 gold, max XP spend, level 1 crafter, etc.)
- [] UI polish (colors, spacing, tooltip formatting)
- [] Error message polish (clear, player-friendly language)
- [] Combat log entries for crafting (“Aldric spent 12 days crafting Wand of Fireball”)
- [] Update Player Guide with crafting section
- [] Update FeatDefinitions descriptions (remove [PLACEHOLDER])
- [] Performance testing (large item lists)
- [] Save/load verification (crafting state persists)

Acceptance: All feats fully functional; no crashes; clear UI; documentation updated.

Phase Summary

Phase	Description	Estimated Days	Priority
1	Core Infrastructure	3-4	 Critical
2	Feat Activation	1-2	 Critical
3	Registry — Database Items	3-4	 Critical
4	Registry — Dynamic Items	2-3	 Critical
5	CraftingExecutor	2-3	 Critical
6	Workshop UI — Main	3-4	 High
7	Item Browser	2-3	 High
8	Cost Preview & Confirm	2-3	 High
9	Arms & Armor Special	2-3	 High
10	Polish & Testing	2-3	 High
	TOTAL	22-32 days	

17. Testing Requirements

17.1 Unit Tests

Cost Calculator Tests

```

public static class CraftingCostTests
{
    // Scroll costs
    [Test] ScrollOfMagicMissile_CL1_Costs12gp_1xp()           // 1×1×25 = 25 → 12gp, 1xp
    [Test] ScrollOfFireball_CL5_Costs187gp_15xp()           // 3×5×25 = 375 → 187gp,
15xp
    [Test] ScrollOfWish_CL17_Costs1912gp_153xp()           // 9×17×25 = 3825 → 1912gp, 153xp
    [Test] ScrollOf0thLevel_Uses0_5Multiplier()           // 0.5×1×25 = 12gp

    // Potion costs
    [Test] PotionOfCureLightWounds_CL1_Costs25gp()         // 1×1×50 = 50 → 25gp
    [Test] PotionOfFly_CL5_Costs375gp()                   // 3×5×50 = 750 → 375gp

    // Wand costs
    [Test] WandOfCureLightWounds_CL1_Costs375gp()         // 1×1×750 = 750 → 375gp
    [Test] WandOfFireball_CL5_Costs5625gp()              // 3×5×750 = 11250 → 5625gp

    // Arms & Armor costs
    [Test] Plus1Weapon_Costs1000gp()                      // 12×2000 = 2000 → 1000gp
    [Test] Plus5Weapon_Costs25000gp()                    // 52×2000 = 50000 →
25000gp
    [Test] Plus1Armor_Costs500gp()                       // 12×1000 = 1000 → 500gp
    [Test] UpgradeWeaponPlus1ToPlus3_Costs8000gp()       // (9-1)×2000 = 16000 →
8000gp

    // Time calculations
    [Test] ScrollOfFireball_Takes1Day()                   // 375/1000 = 1 (minimum)
    [Test] WandOfFireball_Takes12Days()                   // 11250/1000 = 12 (ceil)
    [Test] Plus5Weapon_Takes50Days()                     // 50000/1000 = 50
    [Test] PotionAlways1Day()                             // Special rule
}

```

Validation Tests

```

public static class CraftingValidationTests
{
    // Feat checks
    [Test] CrafterWithoutFeat_Fails()
    [Test] CrafterWithFeat_Passes()
    [Test] AllCastersGetScribeScroll_Free()

    // Caster level checks
    [Test] CL3_CanBrewPotion()
    [Test] CL2_CannotBrewPotion()
    [Test] CL5_CanCraftWand()
    [Test] CL4_CannotCraftWand()
    [Test] CL12_CanForgeRing()
    [Test] CL11_CannotForgeRing()

    // Spell prerequisites
    [Test] KnowsRequiredSpell_Passes()
    [Test] MissingSpell_ShowsSubstitutionDC()
    [Test] PartyMemberHasSpell_Passes()

    // XP safety
    [Test] SufficientXP_Passes()
    [Test] InsufficientXP_Fails()
    [Test] WouldCauseLevelLoss_Fails()
    [Test] ExactlyAtLevelThreshold_Fails()

    // Gold checks
    [Test] SufficientGold_Passes()
    [Test] InsufficientGold_Fails()

    // Arms & Armor special
    [Test] HasMasterworkBase_Passes()
    [Test] NoMasterworkBase_Fails()
    [Test] TotalBonusExceeds10_Fails()
    [Test] UpgradeFromPlus1ToPlus2_Passes()

    // Potion restrictions
    [Test] PersonalSpell_InvalidForPotion()
    [Test] AreaOnlySpell_InvalidForPotion()
    [Test] Level4Spell_InvalidForPotion()

    // Wand restrictions
    [Test] Level5Spell_InvalidForWand()

    // Metamagic rod special
    [Test] MetamagicRod_RequiresMetamagicFeat()
    [Test] MetamagicRod_WithoutFeat_Fails()
}

```

Execution Tests

```
public static class CraftingExecutionTests
{
    [Test] Execute_DeductsGold()
    [Test] Execute_DeductsXP()
    [Test] Execute_CreatesItem()
    [Test] Execute_AddsToInventory()
    [Test] Execute_AdvancesTime()
    [Test] Execute_TriggersResets()
    [Test] Execute_ConsumesBase_ArmsArmor()
    [Test] Execute_TracksHistory()
    [Test] Execute_SetsCraftedMetadata()
}
```

17.2 Integration Tests

Test Scenario	Feats Tested	Verification
Wizard crafts Scroll of Fireball	Scribe Scroll	Scroll in inventory, 187 gp deducted, 15 XP deducted, 1 day passed
Cleric brews Potion of Cure Light	Brew Potion	Potion in inventory, 25 gp deducted, 2 XP deducted, 1 day passed
Wizard crafts Wand of Magic Missile	Craft Wand	Wand (50 charges) in inventory, 375 gp deducted, 30 XP deducted, 1 day
Fighter+Wizard enchants +1 Longsword	Craft Arms	Masterwork consumed, +1 longsword in inventory, 1000 gp, 80 XP, 2 days
Wizard upgrades +1 to +3 Sword	Craft Arms	Old +1 consumed, +3 in inventory, cost is difference only
Wizard adds Flaming to +1 Sword	Craft Arms	Sword gains Flaming, +2 equiv total, checks Fireball prerequisite
Wizard crafts Cloak of Resistance +1	Craft Wondrous	Cloak in inventory, 500 gp, 40 XP, 1 day, checks Resistance spell
Wizard crafts Lesser Rod of Extend	Craft Rod	Rod in inventory, checks Extend Spell feat
Wizard crafts Staff of Fire	Craft Staff	Staff (50 charges) in inventory, checks Burning Hands + Fireball + Wall of Fire
Wizard forges Ring of Protection +1	Forge Ring	Ring in inventory, 1000 gp, 80 XP, 2 days, checks Shield of Faith
Crafter with 0 gold fails	Any	Validation rejects, no deductions
Crafter at XP floor fails	Any	Validation rejects with level-loss warning
Non-caster can't craft	Any	No creation feats available in workshop

17.3 UI Tests

Test	Expected Result
Open Workshop with no crafters	Button grayed out, “(No crafters)” label
Select character updates feat tabs	Only feats the character has are enabled
Switch between feat tabs	Item list updates to correct category
Search filters items	Only matching items shown
Select item shows cost preview	All costs, prereqs, warnings displayed
Insufficient resources disables button	“Begin Crafting” grayed out, reason shown
Confirmation dialog shows correct totals	Gold/XP/time match preview
After crafting, resources updated	Gold, XP, and inventory reflect changes
Multiple crafting sessions work	Can craft several items in sequence
Back to Hub works	Returns to pre-combat hub correctly

Appendix A: Spell Prerequisites for Common Items

Wondrous Items — Top 30

Item	CL	Required Spells
Cloak of Resistance +1-5	5	Resistance
Headband of Intellect +2/+4/+6	8	Fox's Cunning
Amulet of Health +2/+4/+6	8	Bear's Endurance
Gloves of Dexterity +2/+4/+6	8	Cat's Grace
Belt of Giant Strength +4/+6	12	Bull's Strength
Periapt of Wisdom +2/+4/+6	8	Owl's Wisdom
Cloak of Charisma +2/+4/+6	8	Eagle's Splendor
Bracers of Armor +1-8	×2	Mage Armor
Amulet of Natural Armor +1-5	×3	Barkskin
Boots of Speed	10	Haste
Boots of Striding and Spring- ing	8	Longstrider
Bag of Holding I-IV	9	Secret Chest
Handy Haversack	9	Secret Chest
Goggles of Night	3	Darkvision
Eyes of the Eagle	3	Clairaudience/Clairvoyance
Slippers of Spider Climbing	4	Spider Climb
Winged Boots	8	Fly
Pearl of Power (1st-9th)	×2	None (but creator must cast spells of the pearl's level)
loun Stone (Dusty Rose)	12	None
loun Stone (Pale Blue)	12	None
Carpet of Flying	10	Overland Flight
Portable Hole	12	Plane Shift
Monk's Belt	10	

Item	CL	Required Spells
		Righteous Might or monk levels
Vest/Shirt of Resistance +1-5	5	Resistance
Gauntlets of Ogre Power	6	Bull's Strength

Rings — All Implemented

Ring	CL	Required Spells
Ring of Protection +1-5	5	Shield of Faith
Ring of Feather Falling	1	Feather Fall
Ring of Sustenance	5	Create Food and Water
Ring of Swimming	2	None
Ring of Climbing	5	None
Ring of Jumping	2	Jump
Ring of Counterspells	11	Imbue with Spell Ability
Ring of Mind Shielding	3	Nondetection
Ring of Force Shield	9	Wall of Force
Ring of Freedom of Movement	7	Freedom of Movement
Ring of Evasion	7	Jump
Ring of Invisibility	3	Invisibility
Ring of Blinking	7	Blink
Ring of Telekinesis	9	Telekinesis
Ring of the Ram	9	Bull's Strength, Telekinesis
Ring of X-Ray Vision	6	True Seeing
Ring of Spell Turning	13	Spell Turning
Ring of Shooting Stars	12	Light, Faerie Fire, Produce Flame, Lightning Bolt

Appendix B: Caster Level Quick Reference

Class	CL Calculation	Notes
Wizard	= Wizard level	Full caster
Sorcerer	= Sorcerer level	Full caster
Cleric	= Cleric level	Full caster
Druid	= Druid level	Full caster
Bard	= Bard level	Full caster (for CL purposes)
Paladin	= Paladin level – 3	Half caster; CL 1 at level 4
Ranger	= Ranger level – 3	Half caster; CL 1 at level 4
Multiclass	= Highest single-class CL	D&D 3.5e uses highest CL

Appendix C: Maximum Crafting Budget by Level

Level	XP Total	XP Floor	Max XP Spend	Typical Gold
3	3,000+	3,000	~0-3,000	2,700 gp
5	10,000+	10,000	~0-10,000	9,000 gp
7	21,000+	21,000	~0-21,000	19,000 gp
9	36,000+	36,000	~0-36,000	36,000 gp
12	66,000+	66,000	~0-66,000	88,000 gp
15	105,000+	105,000	~0-105,000	200,000 gp
20	190,000+	190,000	~0-190,000	760,000 gp

Note: “Max XP Spend” assumes the character is exactly at level threshold. Characters with XP above the threshold can spend the surplus. Gold values from DMG Table 5-1 (NPC gear value).

This document is the complete implementation specification for the Magic Item Creation Feats system. All phases are designed to be implementable directly from this plan without additional research.